

Accelerating Domain-Specialized AI Research Through Open Source

Building Reliable, Controllable, and Sustainable AI for Our Domain

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Meet the Speaker



Chansung Park

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- 2012** – Research on intelligence of optical networks at ETRI
 - 2020** – Administrator, AGI KR
 - 2020** – Google Developer Experts – AI
 - 2022** – Google Developer Experts – Cloud
 - 2022** – Hugging Face Fellow – transformers, diffusers, trl, alignment-handbook
 - 2023** – Google for Startups Accelerator – tech mentor for 6 startups globally
 - 2024** – Mistral AI Ambassador
 - 2024** – dstack.ai Ambassador
 - 2025** – Member of the Subcommittee, National AI Strategy Committee in Korea
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01

AI Is Everywhere



The ChatGPT Moment

AI is everywhere.



Daily Life



Work & Productivity



Education



Healthcare



Finance



Creative Work

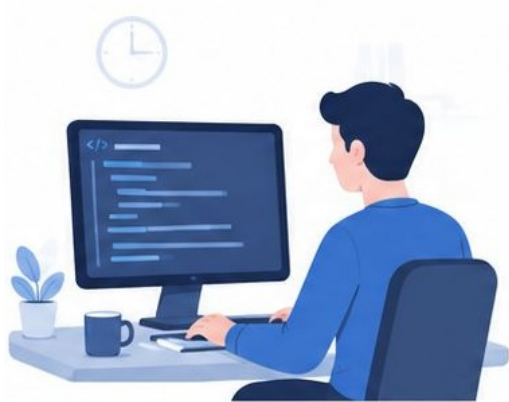
Across industries and professions, AI has been rapidly adopted.



The Agentic Coding Shock

Code productivity has skyrocketed.

Before



- Manual coding
- Repetitive tasks
- Limited by time



Now



- AI pair programming
- Automated workflows
- Beyond what we thought possible

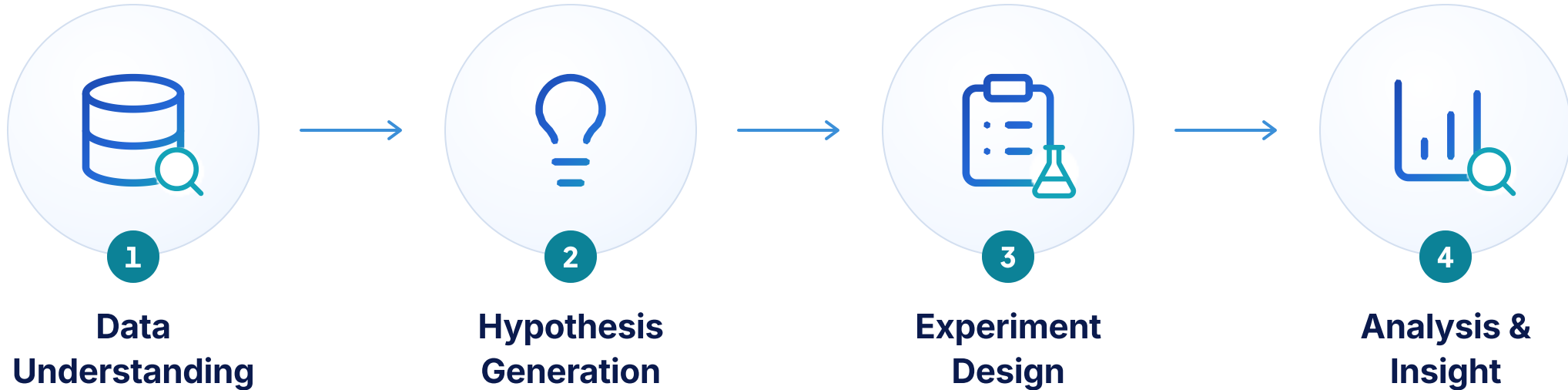


We can now build things we couldn't before.



Research Is Changing Too

AI is introducing a new paradigm in domain research.

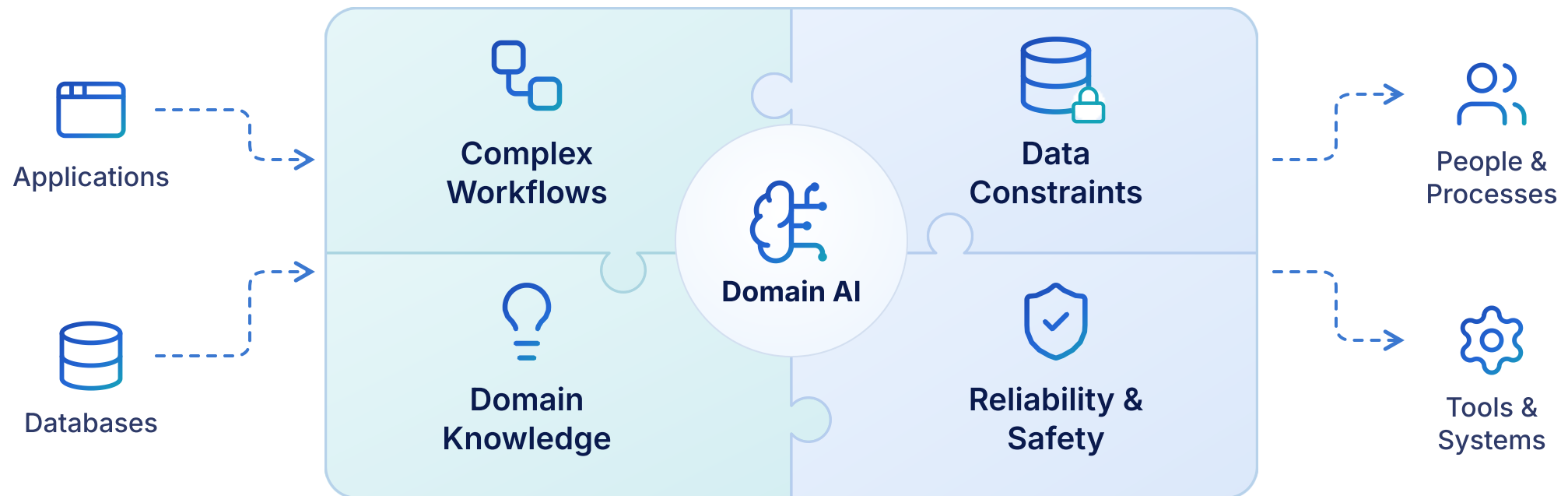


From narrow tools to research partners.



But Domain AI Is Hard

It's not just about calling an API. It requires deep integration with existing systems and workflows.



🎯 The challenge is not only intelligence, but integration.



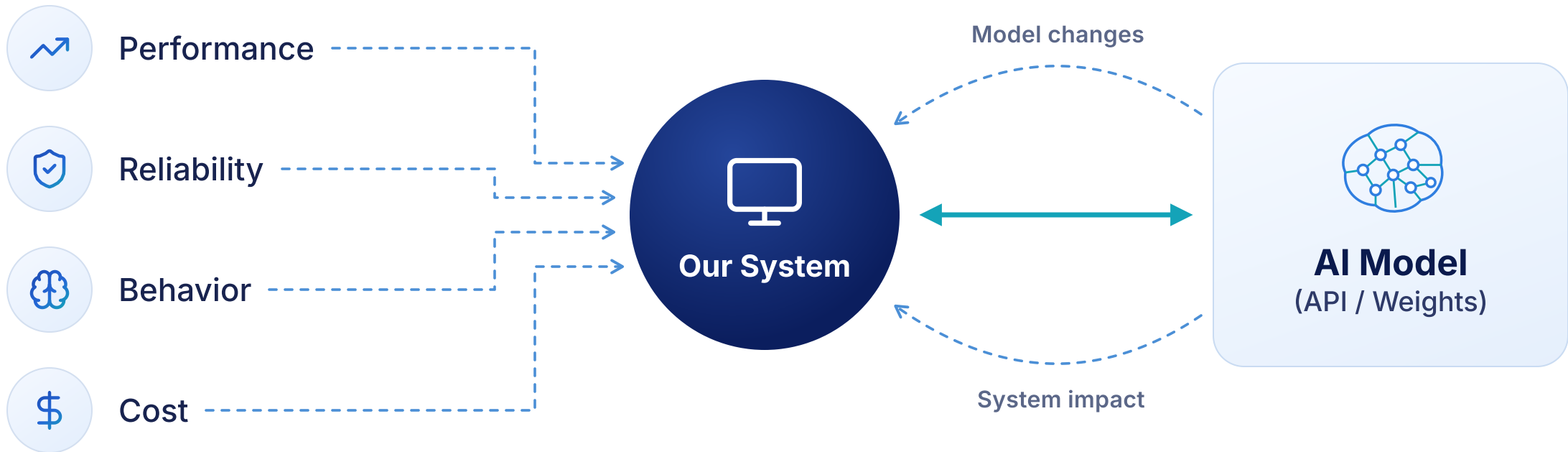
02

Model Dependency & Drift



AI Model Dependency

Domain applications become highly dependent on specific models.



A strong application can still be fragile if the model shifts.



Models Are Not Under Our Control

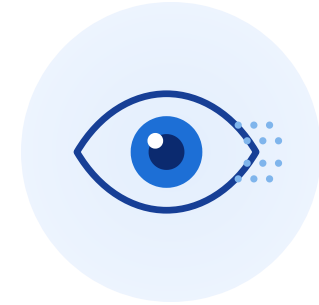
API models change due to updates, policies, and regulations.



Performance can
drop without notice



Usage restrictions
can be imposed



Behavior can
change silently

We can't fully control what happens.

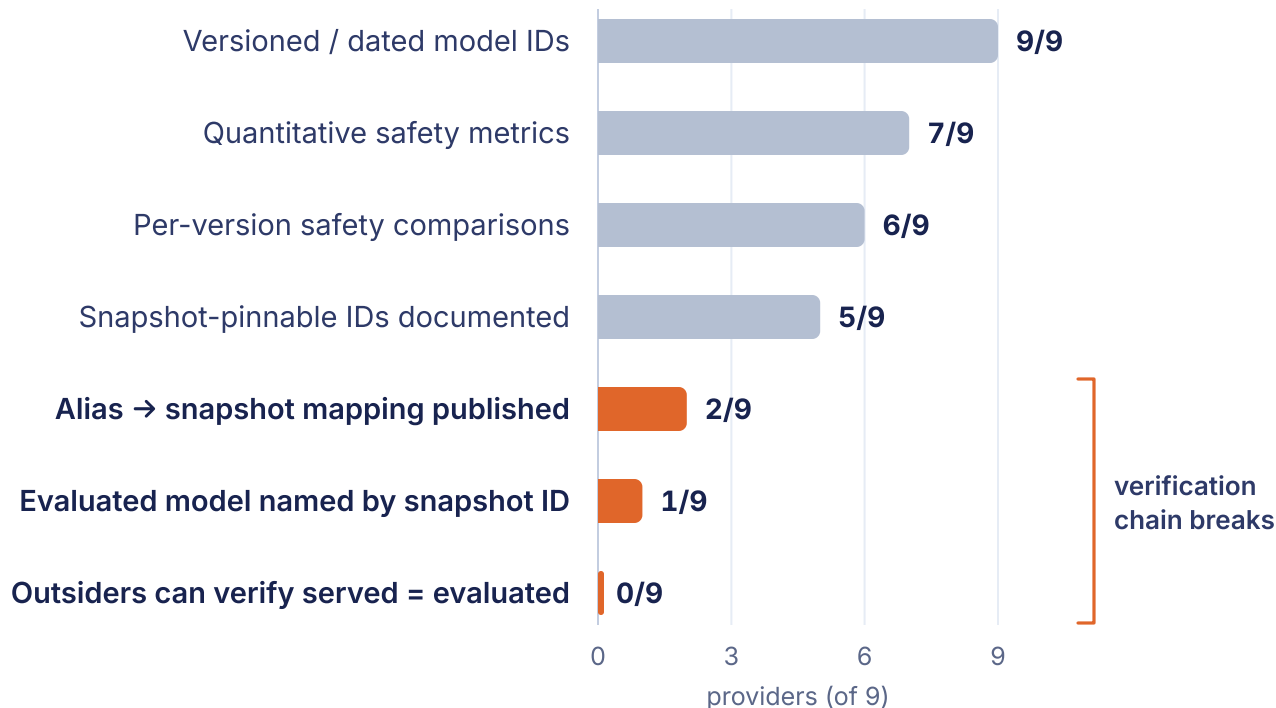


A Model Name Is Not a Frozen Instrument

What was evaluated is not guaranteed to be what is served — or what answers tomorrow.

Silent updates: what 9 API providers disclose

Providers can change a deployed model without a new version name.



0.78

agreement when **byte-identical requests** to the same model name were replayed — the study required 0.99. Same-day replays were no better (0.805). Four providers: medians 0.74–0.88.

8 of 53

monitored API endpoints **changed behavior within 23 days** — one confirmed: a host silently redirected Mistral-7B to a different model.

Sources: Abraham & Bucknall, arXiv 2608.11803 · Zhu & Zhang, arXiv 2609.04198 · Chauvin et al., arXiv 2602.11083 (all 2026)



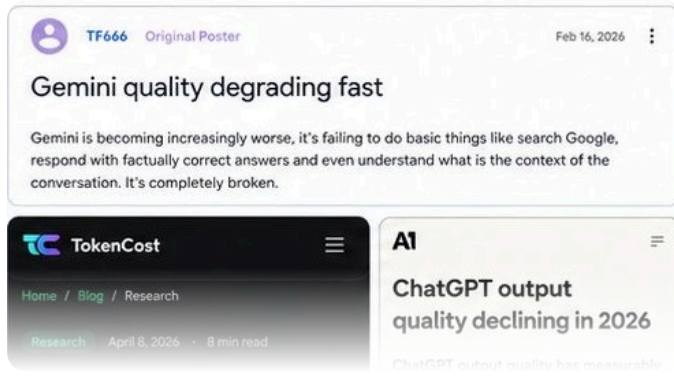
Even Frontier Labs Are Not Immune to Quality Regressions

Signals from users, vendors, and media suggest that output quality can shift over time.



User Reports

- Community posts report quality drops in real usage
- Symptoms include weaker reasoning and unstable behavior
- Even simple tasks can feel less reliable



Vendor Acknowledgements

- Providers sometimes investigate or explain regressions
- Quality changes may come from updates around the model
- The API or product experience can shift over time



Media Coverage

- Worsening output quality becomes visible enough to make headlines
- Token limits, product changes, and regressions affect perception
- The key issue is that the end user cannot fully control this



Hosted models are powerful, but they are also moving targets.



The Model Didn't Change. The Agent Did.

Weeks of "Claude got worse" reports traced to effort defaults, context handling, and one prompt line.

Claude Code, March–April 2026: three product-layer changes

Model weights and the API were not affected.

Default reasoning effort lowered

high → medium

Sonnet 4.6 · Opus 4.6

Old thinking cleared on every turn

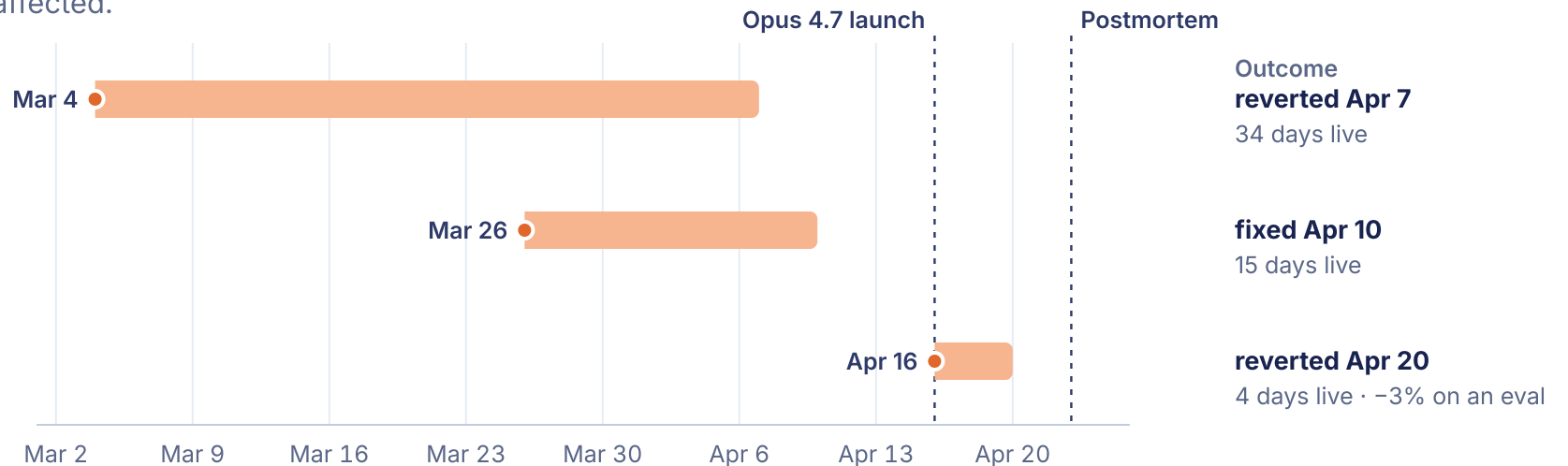
caching change with a bug

Sonnet 4.6 · Opus 4.6

System-prompt length cap

≤25 words between tool calls

Sonnet 4.6 · Opus 4.6 · Opus 4.7



Each change hit a different slice of traffic on its own schedule — it looked like random degradation.



The caching bug passed code review, tests, and dogfooding; the prompt line, weeks of evals.



Internal evals did not reproduce it at first — user reports led to the root causes.

Source: Anthropic, [An update on recent Claude Code quality reports](#), engineering blog (Apr 23, 2026)



Context Drift: When Intent Evolves, Models Get Lost

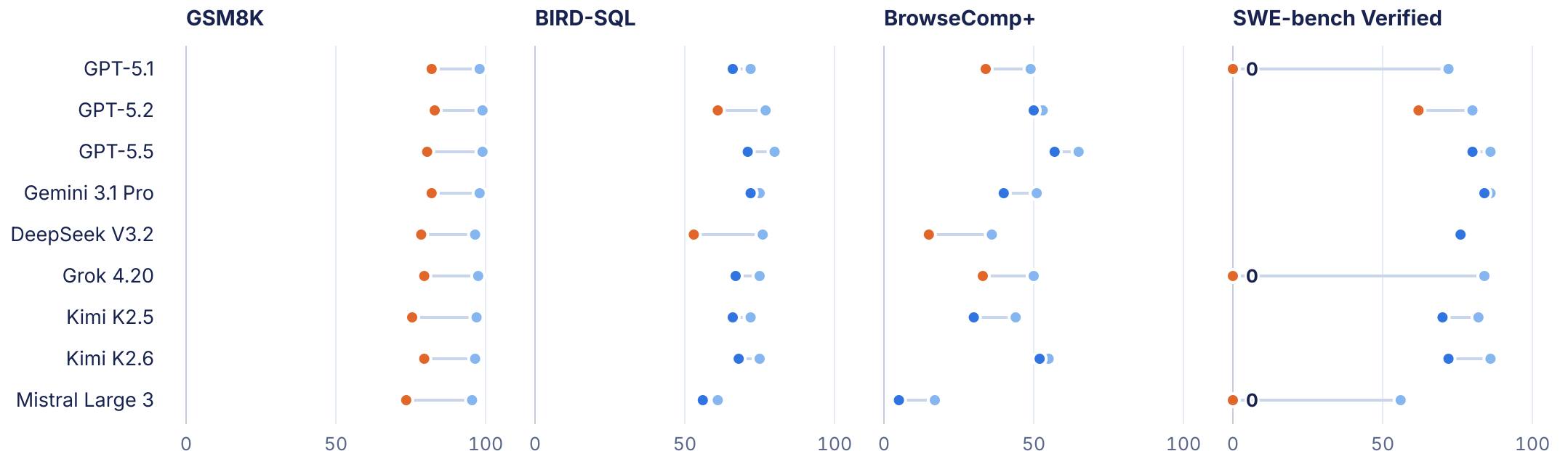
Strong single-turn scores do not survive a conversation in which the goal keeps changing.

Accuracy (%) — same tasks, intent revealed, revised and redirected over 7 turns

● Single turn, fully specified

● Evolving intent, 7 turns

● Evolving intent, drop ≥ 15 pts



i The drift, not the length: same 7 turns — 3 of them repeats → 87.5%; 3 more intent changes instead → 82.0% (GPT-5.1, GSM8K).

Source: Tack, Laban & Neville (Microsoft Research), **LLMs Get Lost in Evolving User Intent**, arXiv 2607.20734 (Jul 2026), Tables 1 and 5



03

The Migration Treadmill

03

What About Open-Weight Models?

We can get a model snapshot and host it locally.
That gives us stability—at least for a while.

- ✓ | Own the weights
- ✓ | Run on-premise
- ✓ | No surprise policy changes
- ✓ | Reproducible at a point in time

! But the world keeps moving.





Inevitable Model Migration

AI technology is evolving fast. Stronger models keep emerging.



Can we ignore a much better model without falling behind?

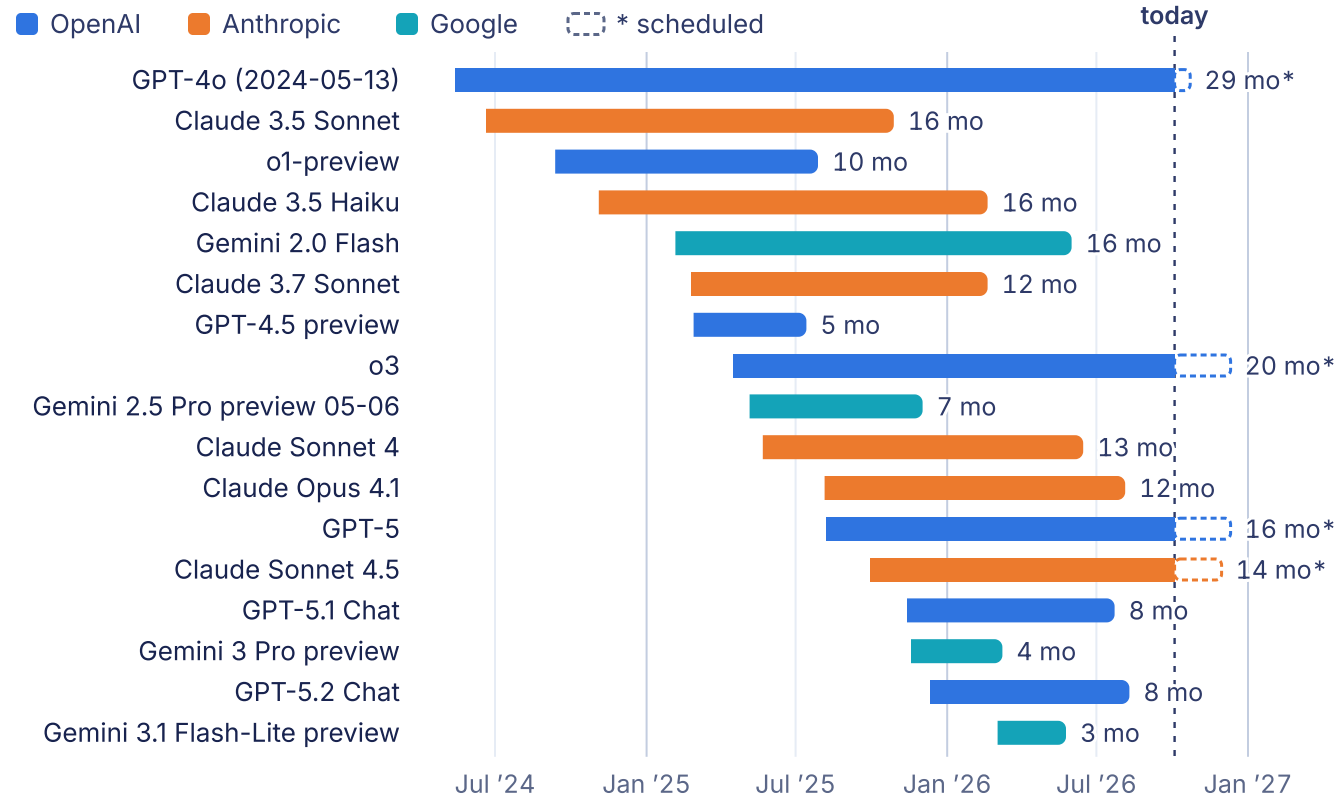


Models Expire — and Teams Migrate Too Late

API models are retired on the vendor's schedule, often about a year after release.

API models: release → retirement

Median lifespan (actual or scheduled) of these 17 models: 12 months



82%

of migrations off retired models were committed **after the shutdown date** — a median of 39 days late.

94%

of migrating apps had the **model ID hard-coded**.

4.5%

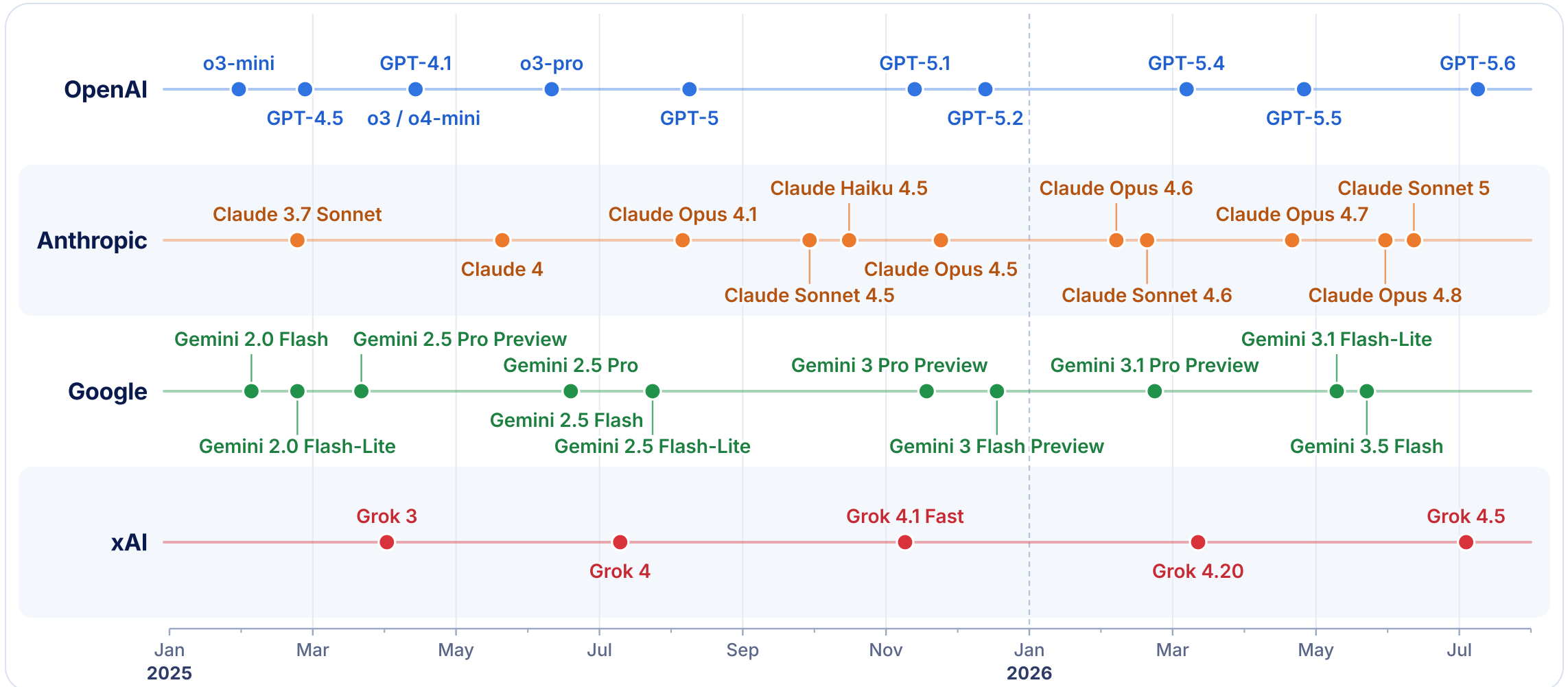
of migration commits mention an **evaluation or regression suite**.

Sources: OpenAI · Anthropic · Google deprecation pages (Oct 2026) · H. L. Kim, **When the Model Retires**, arXiv 2609.31288 (Sep 2026)



The Frontier Model Layer Keeps Moving

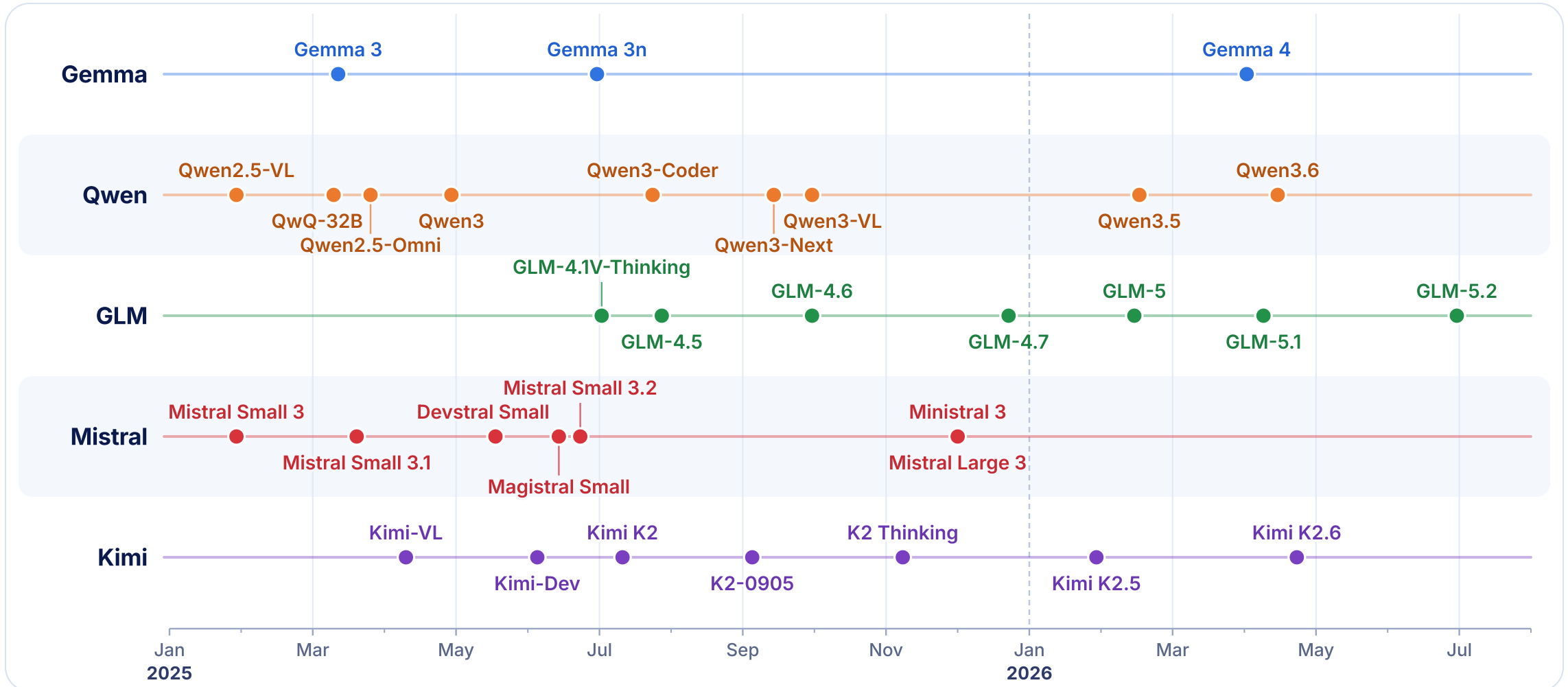
Major hosted-model releases from Jan 2025 to Jul 2026





Open Models Keep Coming — by Design

Openness is a strategic advantage. Selected open-weight families, Jan 2025–Jul 2026



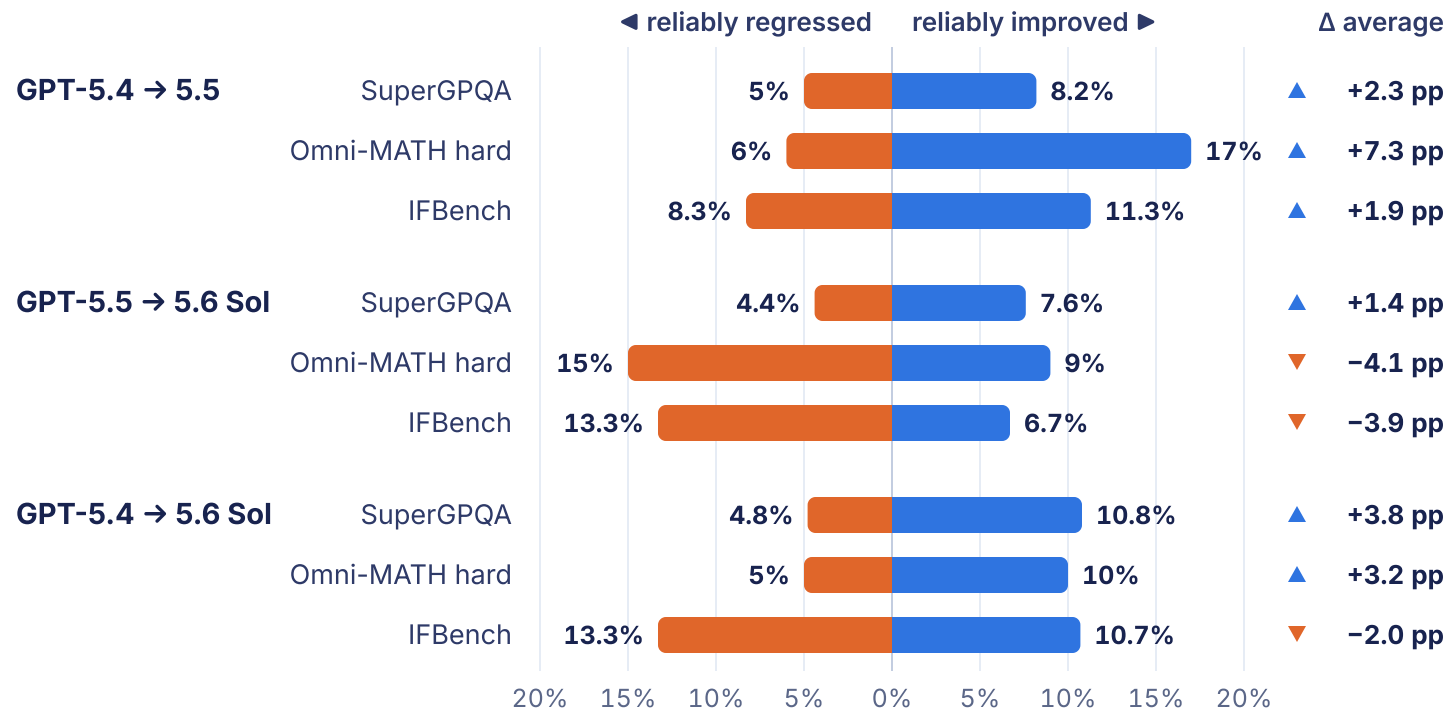


Even When the Average Rises, Upgrades Break Things

Reliable improvements and reliable regressions coexist in every upgrade measured.

Item-level change on GPT upgrades

900 benchmark items, each sampled 50× per model (FDR-controlled, ≥ 20 pp shift)



+7.3 pp

average gain on Omni-MATH hard from
GPT-5.4 → 5.5 ...

6%

... yet **6% of those items** got reliably worse.

15%

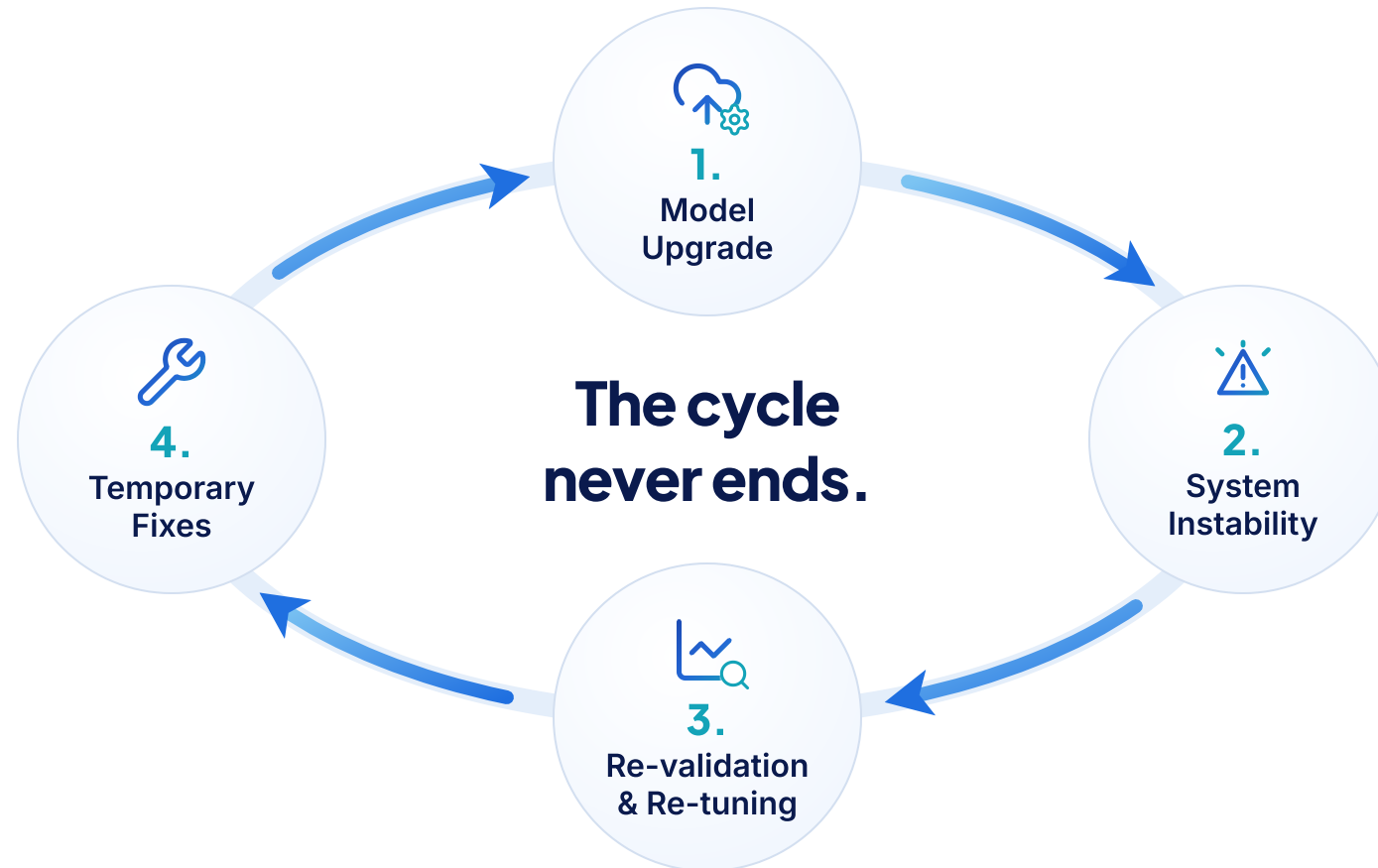
of Omni-MATH hard items **regressed** on
the next upgrade (5.5 → 5.6 Sol).

Source: Xu & Wu, **What Aggregate Scores Miss**, arXiv 2608.17719 (Aug 2026), Table 1 — GPT-5.4, GPT-5.5, GPT-5.6 Sol via the OpenAI API



The Real Problem

Upgrading models means our system becomes unstable again.





What We Need to Do

To make Domain AI work, we must:



Secure our domain data



Build controllable AI systems



Continuously adapt and improve



Control, not just adoption.



04

Why Open Source



Can We Build Giant Models Ourselves?

Training frontier models from scratch requires:



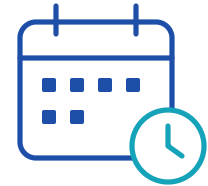
Massive Compute



Top-tier Talent



Huge Investment



Long-term Commitment

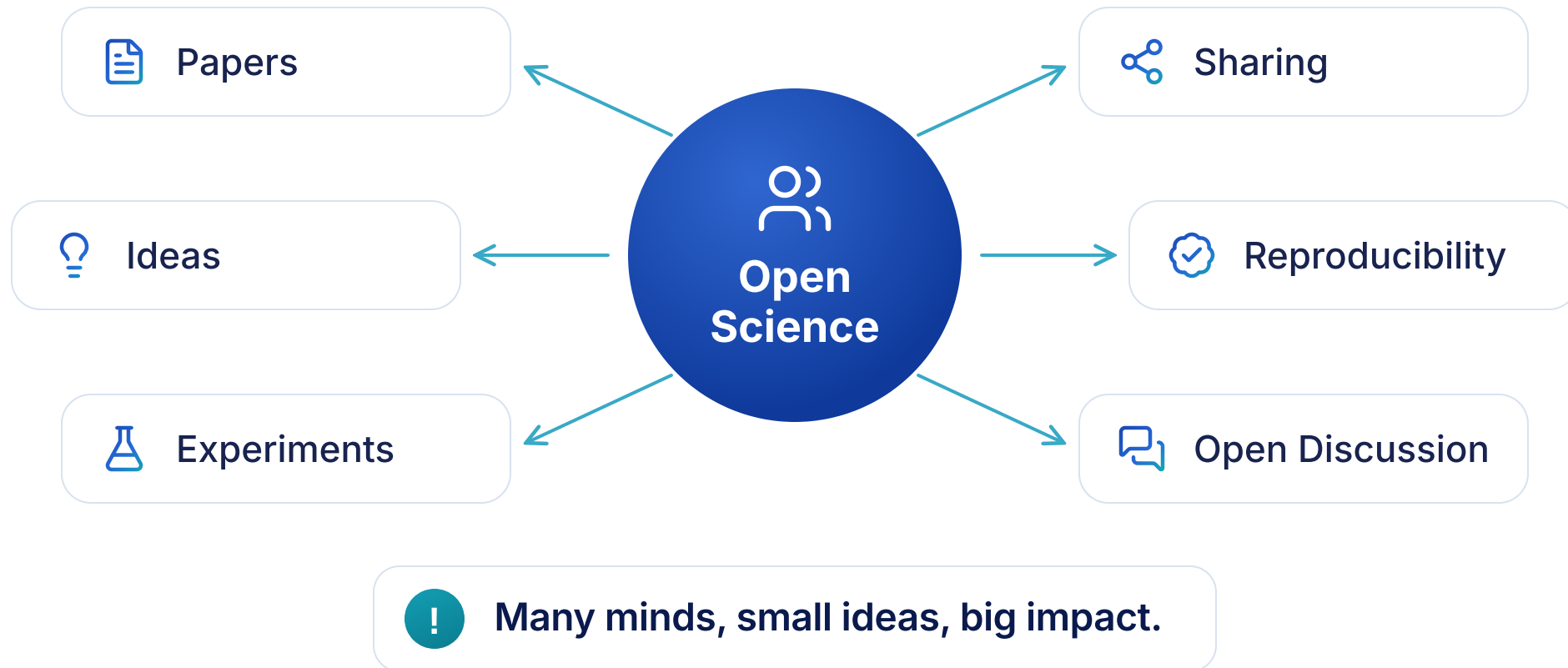


Only a few companies or institutions can play this game.



Open Science Drives AI Forward

AI breakthroughs cannot be led in secret by a single company.





Why Open Models Keep Coming

Organizations open-source their models and code to:

-  Increase technical influence
-  Attract top talent
-  Encourage contributions
-  Grow the ecosystem
-  Accelerate real-world adoption

 **Openness is a strategic advantage.**





Back to the Main Question

Assuming we have secured our domain data,



What should we do next?

Let's focus on what comes after data.



What Is Open Source?

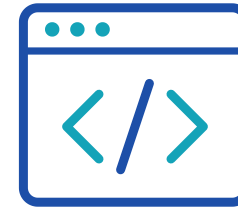
"Open" and "Source"

Open



Open to everyone

Source



Access to the source code

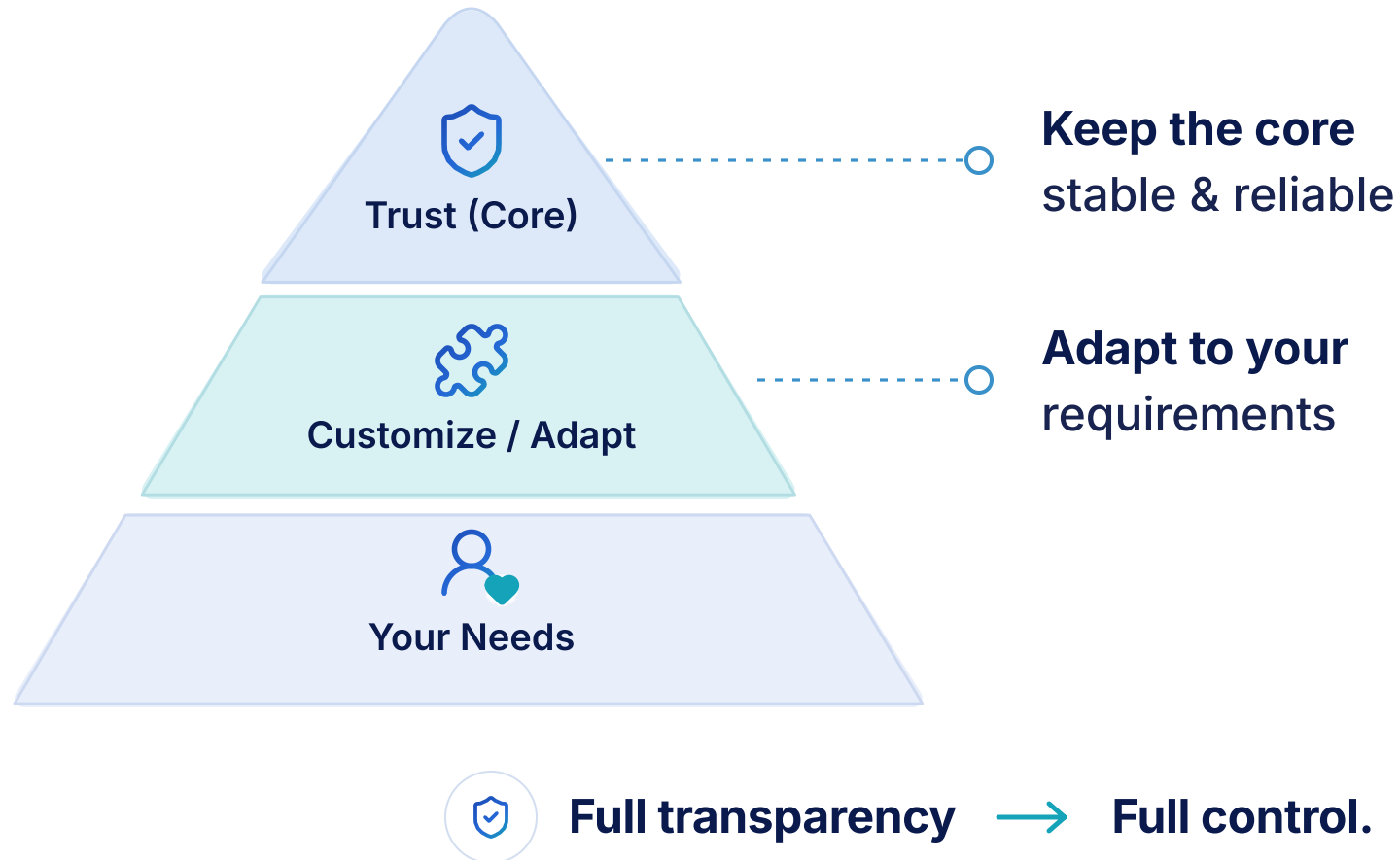


Freedom to use, understand, modify, and share.



The True Power of Open Source

You can trust the core, and customize the rest.





Open-Source AI Is No Different

We get excited about:



New open-weight model releases



Running models on our own infrastructure



Building agents with more control



Reducing dependency on closed APIs



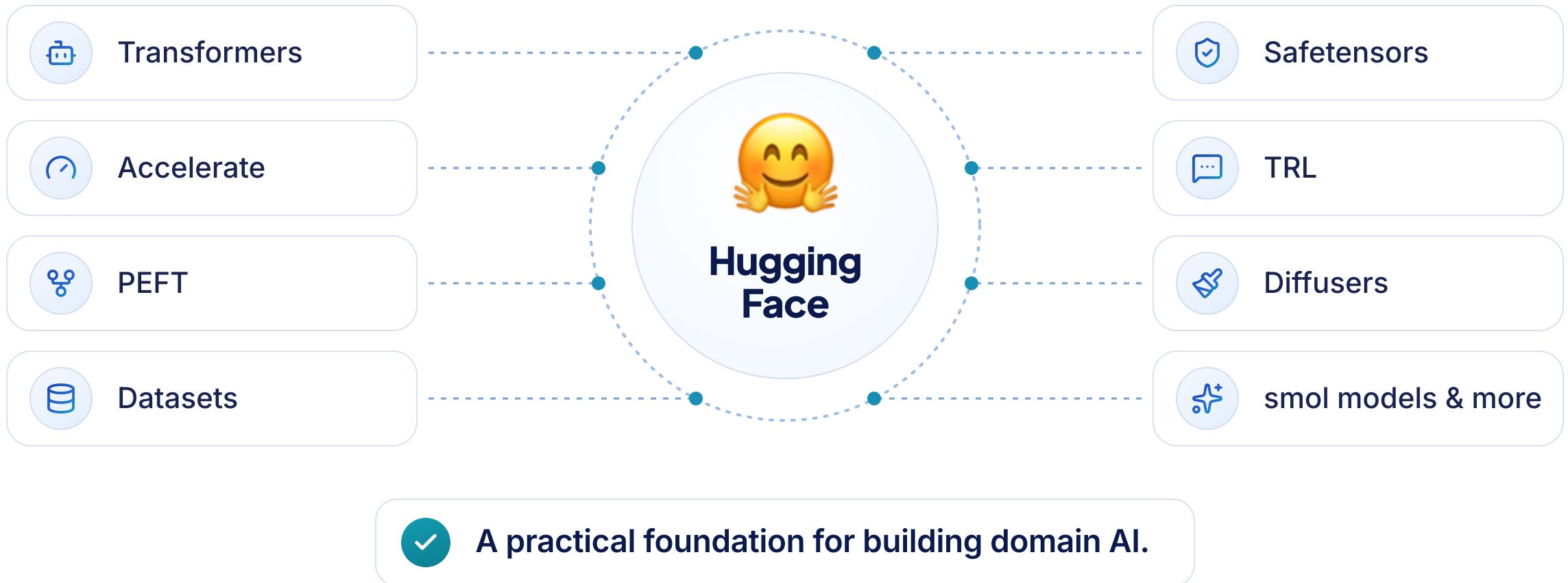
But using them “as-is” is just the beginning.





The Hugging Face Ecosystem

A massive open-source ecosystem powering AI development.





Research Meets Engineering

The open-source AI ecosystem turns research ideas into usable tools.



Good infrastructure connects innovation to reality.



05

Toward Reliable Network AI

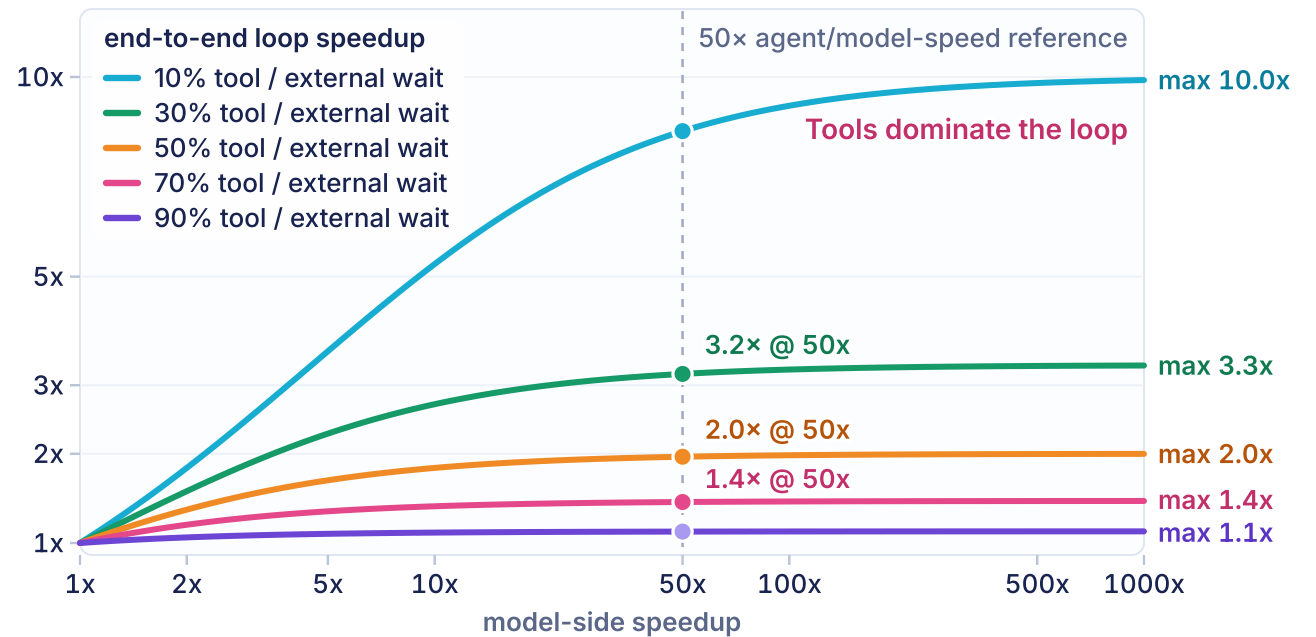
05 Rapid AI Progress Makes the Future Harder to Predict



A major bottleneck for AI agents is no longer only the model itself. Many surrounding tools and systems were still designed around human-speed workflows.

From Jeff Dean at NVIDIA GTC

Amdahl Wall for AI Agents



A visual interpretation of Amdahl's Law

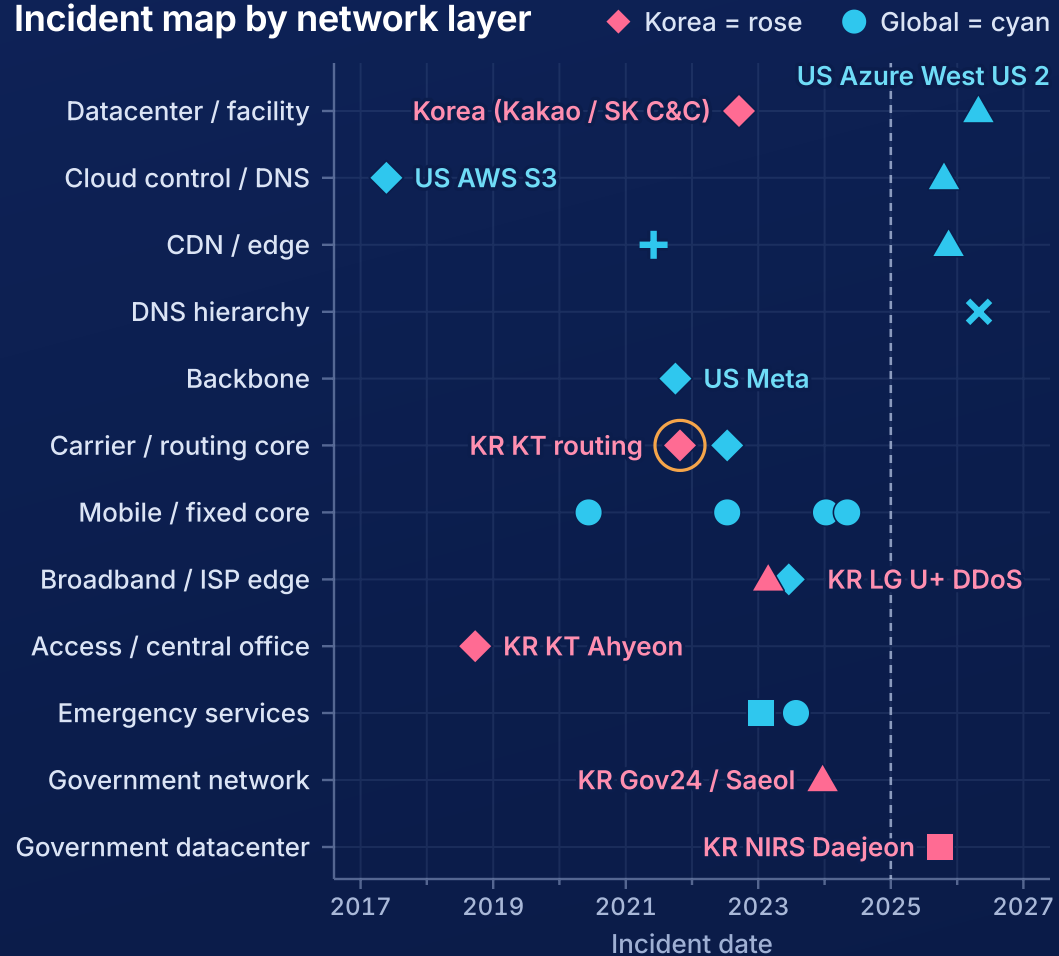
If 90% of agent-loop time is spent on tool calls or external interactions, even a 10x faster model cannot deliver 10x end-to-end improvement.

Amdahl's Law shows that the non-improvable portion creates a hard limit on total system speedup.

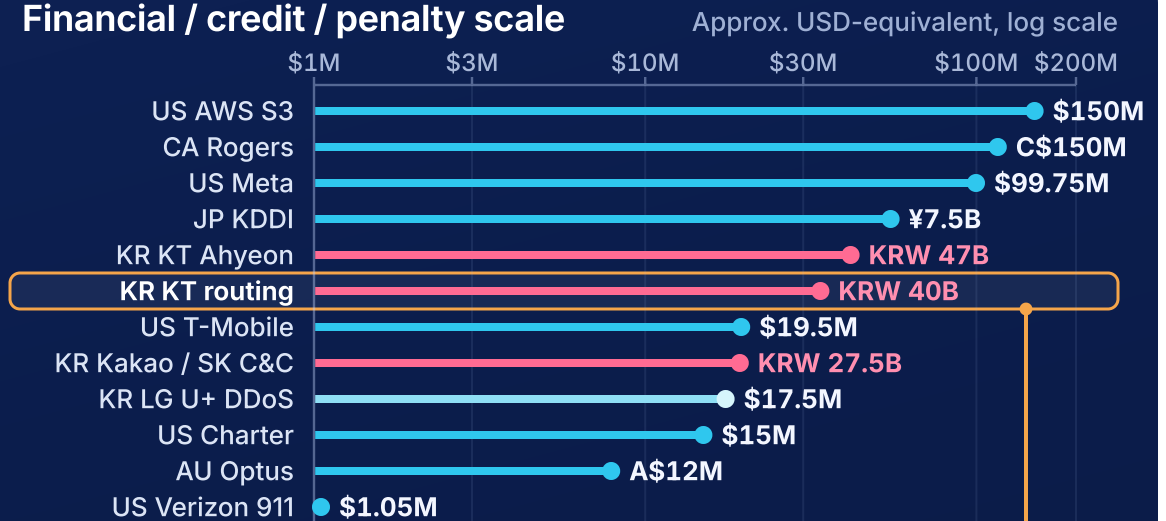
Network Operations Mistakes Can Be Extremely Costly

Global + Korea Network-Ops Blast Radius Evidence

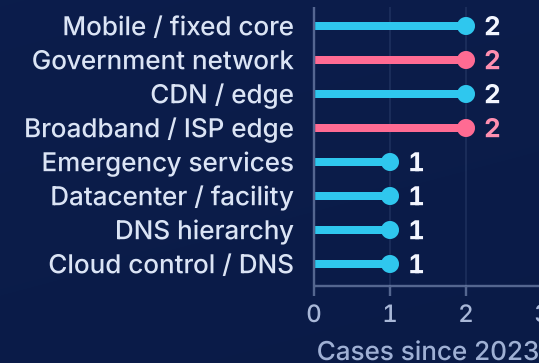
Incident map by network layer



Financial / credit / penalty scale



Recent cases by layer



**2021 KT routing
configuration mistake**

Estimated loss:
more than **KRW 40B**



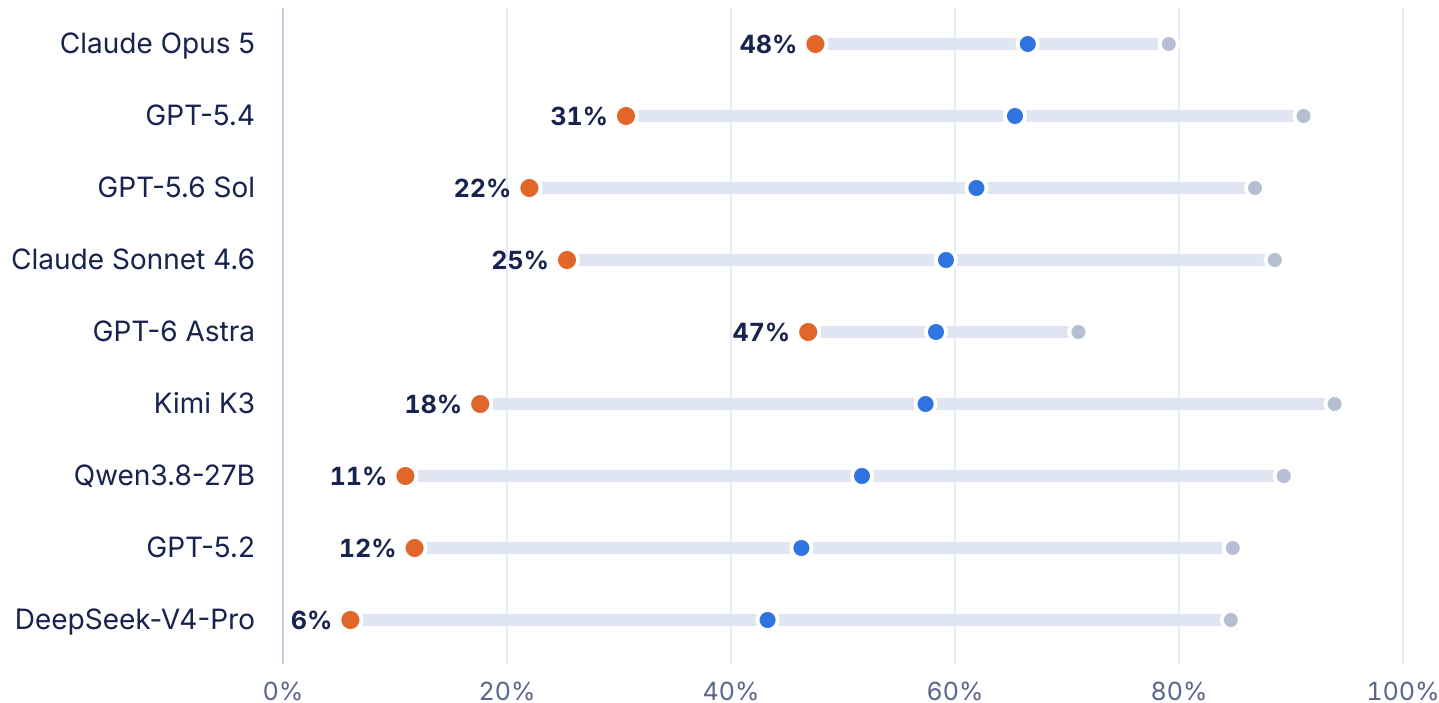
One Success Isn't Reliability

Agents can usually solve a task once. Solving it every time is a different story.

Agents on 507 stateful business workflows, 20 trials each

Share of tasks solved — per trial, every time, at least once

● Every time (pass²⁰) ● Average run (pass@1) ● At least once (pass@20)



66% → 48%

best model (Claude Opus 5): average run vs. **success on all 20 trials**.

65% → 31%

GPT-5.4 — the gap is not a small-model problem.

≈54%

of outcome variance came from **re-running the same agent configuration** (separate study).

Sources: Li et al., **Thinkingbox**, arXiv 2608.19741 v4, Table 15 · Wiedmann et al., arXiv 2610.01618 · both Oct 2026

Compiled Specs Beat Growing Context

A compilation-oriented agent keeps one explicit spec as shared state — demonstrated on a real 25G PON testbed (ECOC 2026).

1 Experimental Network Setup



2 Trace Validation Simulator

Build a digital twin to verify network control traces

3 Open-Weight Agent Systems

Develop and evaluate agent systems with diverse open-weight models

- Gemma
- Qwen
- others

4 Agentic Small Language Model

Use verified traces for small-model training

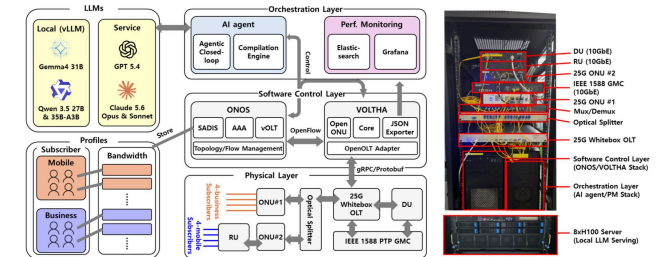


Goal: reliable, controllable, and data-efficient network AI.

- ✓ Highest multi-step success: 80% at S7, vs. 50% and 0%.
- ✓ 0.71x the tokens of accumulated step history (534k vs. 752k).
- ✓ Gains hold across six LLMs — service APIs and open weights.



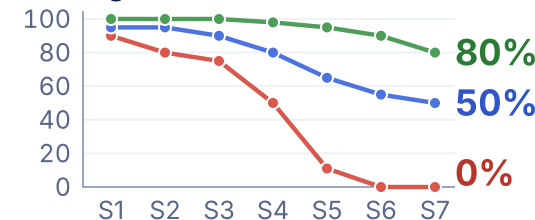
25G PON Testbed (SEBA stack)



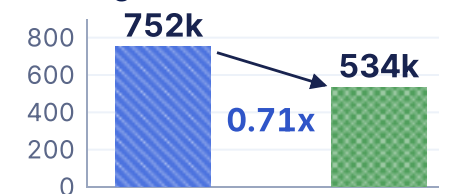
Representative Results

■ Prev-step only ■ Accumulated history ■ Compiled spec (ours)

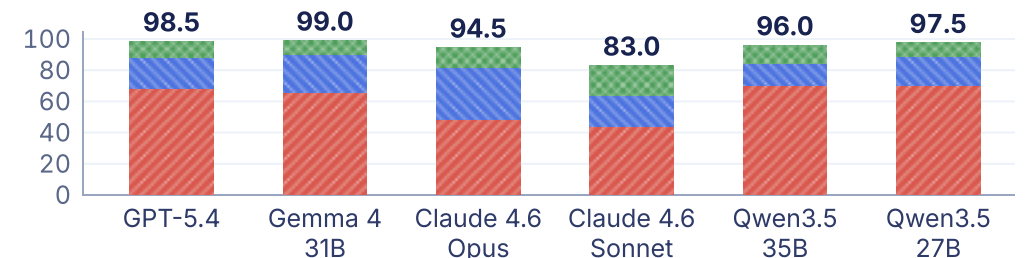
(a) Avg. cumulative success rate (%)



(b) Avg. tokens (K)



(c) Success rate (%)



Verified Traces Beat Bigger Models

Fine-tuned on verified steps, a 2B open model solves every held-out task — base models given the docs reach at most 28%.

1 Collect verified runs



The frontier LLM is used only here — offline.

2 Build step-wise samples



*State-changing commands only · no docs

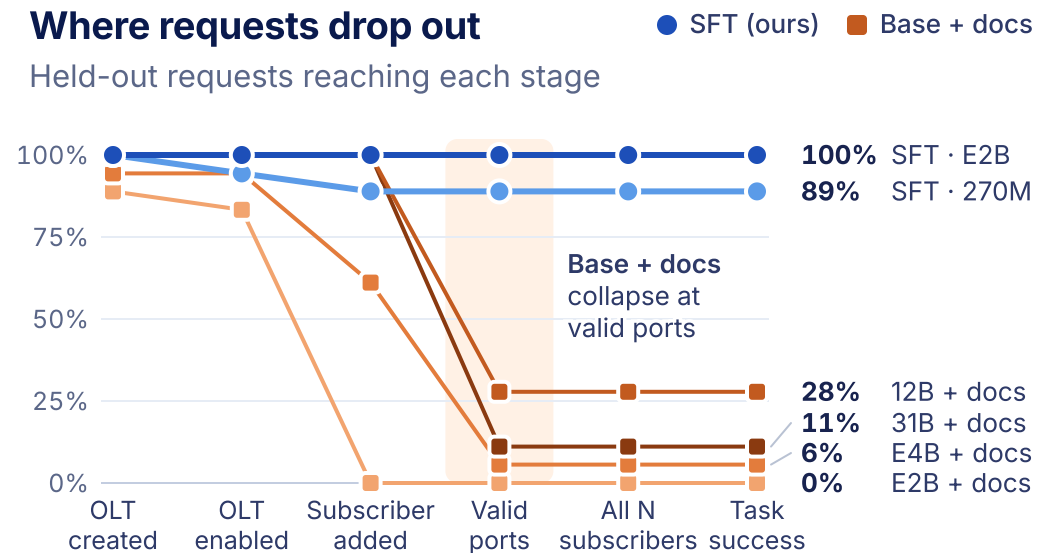
3 Train a small open model



4 unseen intents · no docs

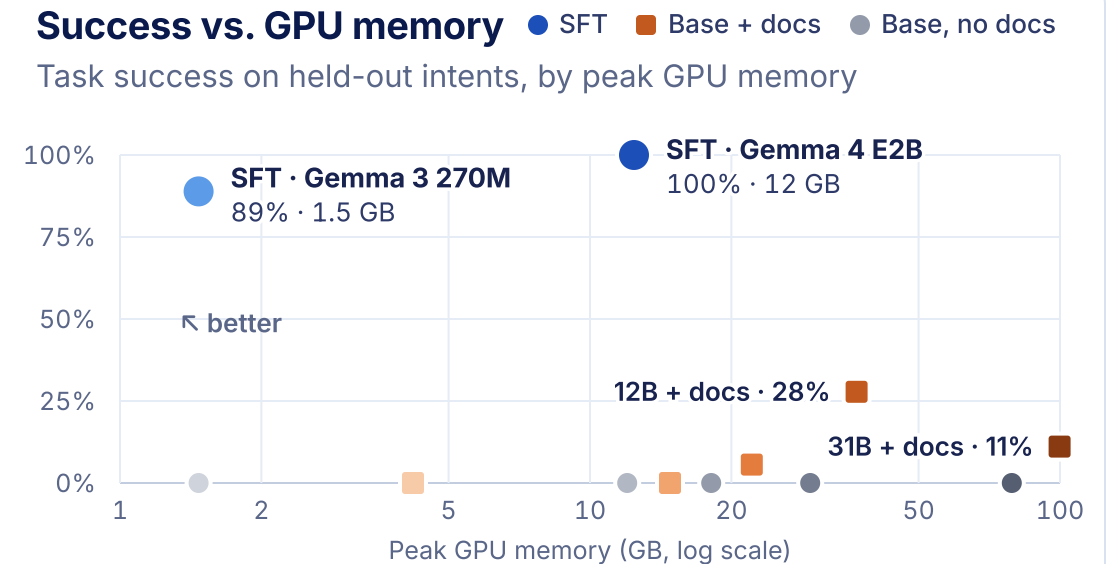
Where requests drop out

Held-out requests reaching each stage



Success vs. GPU memory

Task success on held-out intents, by peak GPU memory





Other Labs, Same Recipe

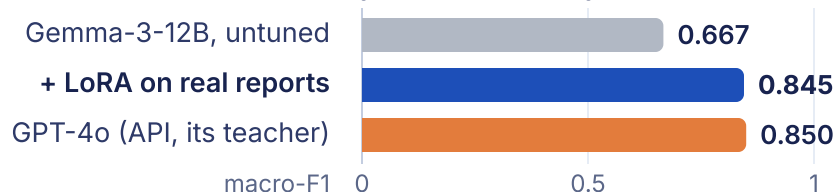
Four 2026 studies: an open model plus the domain's own data matches or beats strong baselines — on infrastructure they control.

■ Open model, untuned or general ■ Open model + domain data ■ Closed API or best public model

MEDICINE Emory University · Sep 2026 · preprint

A 12B open model ties GPT-4o on head-CT reports

Gemma-3-12B + LoRA on 2,000 real in-house reports, one GPU. Real reports beat GPT-4o-written synthetic ones by 0.20 F1.

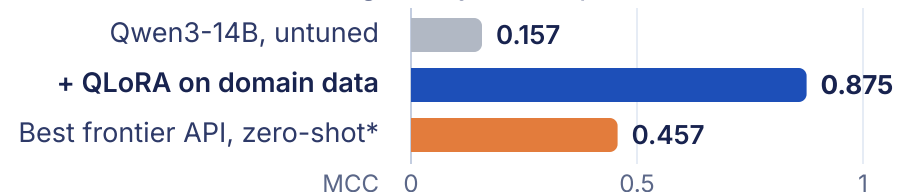


Mansuri et al., arXiv 2609.40236 (2026)

MATERIALS · KOREA Hanyang Univ. & KENTECH · Apr 2026

Same open 14B model: MCC 0.16 → 0.875

QLoRA on 364k curated compositions (Materials Project, OQMD); runs in <15 GB. Goal: "eliminating costly API dependencies."

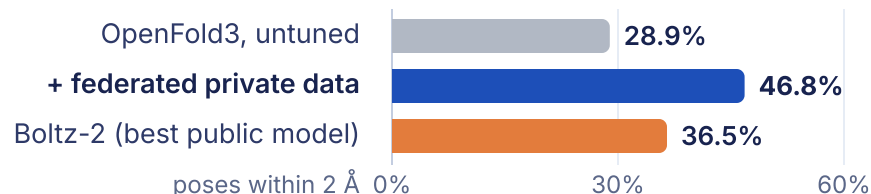


Yoon, Gu & Kim, npj Comput. Mater. 12, 221 (2026) · *zero-shot, on a 10k subset

DRUG DISCOVERY AbbVie, Astex, BMS, J&J, Takeda + Apheris · Sep 2026

Five pharmas tuned an open model without sharing data

Federated fine-tuning of open OpenFold3 on 20,167 private protein-ligand structures.

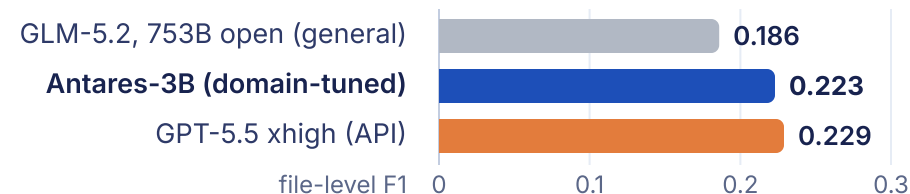


Apheris / AISB Network / AIQuraishi Lab, tech. report (Sep 2026)

SECURITY Cisco Foundation AI · Aug 2026

A 3B model nearly matches GPT-5.5 — at ~1/170 the cost

Granite 4.0 + SFT/RL on verifiable vulnerability tasks; runs air-gapped. \$0.82 vs. \$141 for the 500-task benchmark.

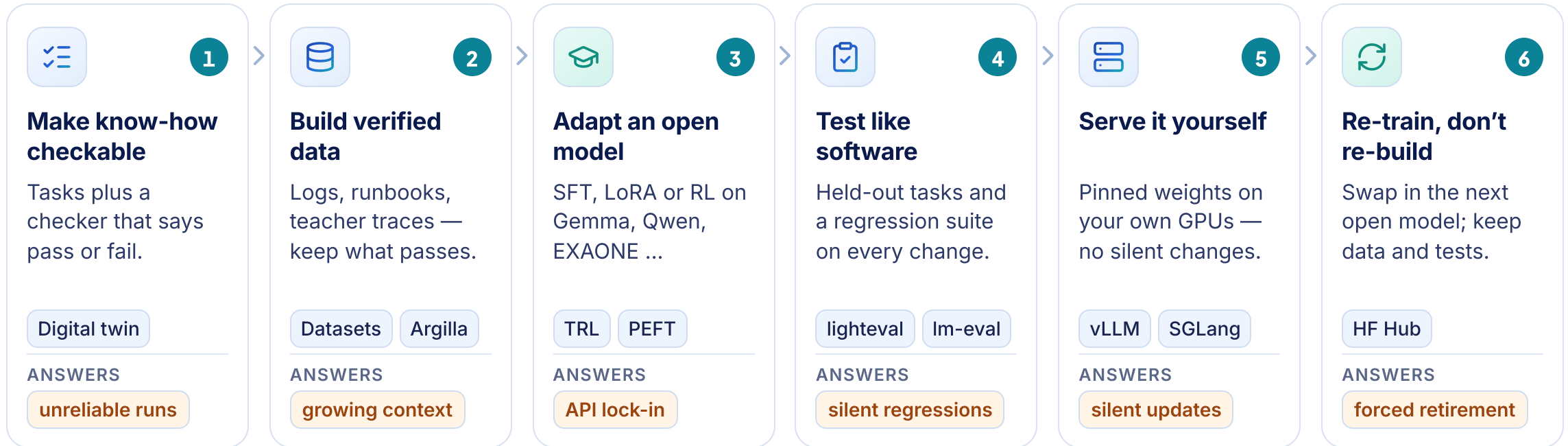


Cisco Foundation AI, arXiv 2608.02407 (2026)



A Recipe for Domain AI on Open Source

Six steps any domain team can run, with open models, open tooling, and its own data.



..... Next open model? Re-run steps 3–5 on the same data and tests.



Models are replaceable. Verified data, checkers, and tests are what you keep.



Key Takeaways

From depending on external models to owning a domain system.

1



Hosted models drift, regress, and retire — systems built on them are fragile.

2



Open models + domain data put the weights, the data, and the tests in your hands.

3



Reliability comes from verified data: our tuned 2B model solved every held-out task.

4



And it lasts: when a better open model arrives, re-train — don't re-build.



From adoption to control.

Q&A

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